Al-Balqa Applied University
Faculty of Engineering Technology
Department of Electrical Engineering

Signals and Systems
Midterm Exam
1st Semester 2023/2024 05-12-2023

Grade: 29

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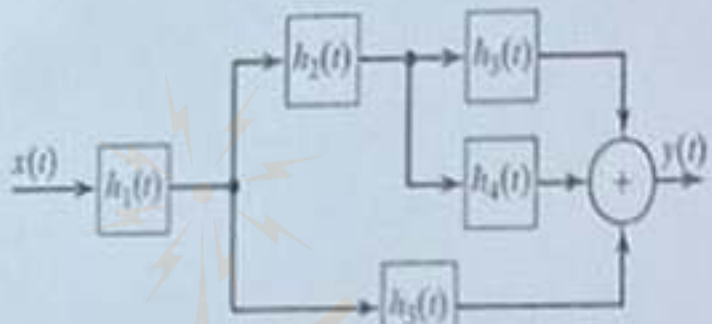
Q1. Select the correct answer for the following questions:

1. Which of the following is not true?

a. $t^2\,\delta(t) = 0$ b. $cos(t)\,\delta(t) = \delta(t)$ c. $sin(t)\delta(t-\pi) = \delta(t-\pi)$ d. $\int_{-\infty}^{\infty}(2t^2+1)\,\delta(t)dt = 1$

2. The total impulse response of the LTI system given below is.....

a. $h(t) = h_1(t) * ([h_2(t) * (h_3(t) + h_4(t))] + h_5(t))$
b. $h(t) = h_5(t) * ([h_2(t) * (h_3(t) + h_4(t))] + h_1(t))$
c. $h(t) = h_1(t) + ([h_2(t) * (h_3(t) * h_4(t))] + h_5(t))$
d. $h(t) = h_2(t) + ([h_1(t) * (h_3(t) + h_4(t))] + h_5(t))$



3. A system is given as $y(t) = cos(x(t)) + 2\,sin(x(t-1))$. This system is described as .....

a. Causal and memory. b. Stable and memory. c. Stable and memoryless. d. Non-causal and memory.

4. If $x(t) = e^{cos(t)}$, then the signal is.....

a. Odd and periodic signal. b. Even and periodic signal. c. Odd signal. d. Neither even nor odd.

5. The result of $\int_{-2}^{6} 5\,\delta(t+4)\,dt$ is given as.....

a. 0 b. 5t c. 5t + 20 d. 5

6. The power of the signal $s(t) = 8\,cos\left(20\pi t - \frac{\pi}{2}\right) + 4sin(15\pi t)$ is.....

a. 12 b. 82 c. 40 d. 64

7. The function x (t) = 6 rec (t / 30) has a value of.....

a. 1 for $-30 \leq t \leq 30$.
b. 6 for $-30 \leq t \leq 30$.
c. 1 for $-15 \leq t \leq 15$.
d. 6 for $-15 \leq t \leq 15$.
e. 1 for $-60 \leq t \leq 60$.
f. 3 for $-60 \leq t \leq 60$.

8. Non-periodic signals are the examples of ..... signals with ..... .

a. Energy, Zero average power b. Power, Zero energy. c. Energy, ∞ average power. d. Power, ∞ energy.

9. The fundamental period of the signal $x(t) = \cos(6t/4 + 40^\circ) + 2\sin(8t/3)$ is:

a. $\pi/4$ b. 12π c. $3\pi/4$ d. $\pi/3$

10. The result of $\cos(2t) + \cos(-2t)$ is .....

a. $0.5\cos(t)$ b. e. $2\cos(t)$ c. $2\cos(4t)$ d. $0.5\cos(2t)$ $2\cos(2t)$

11. Which of the following system is causal?

a. $y[n] = 2x[n] - 3x[n+1]$

b. $y[n] = 2x[$

c. $y[n] = 2x$ $-3x[n+1]$

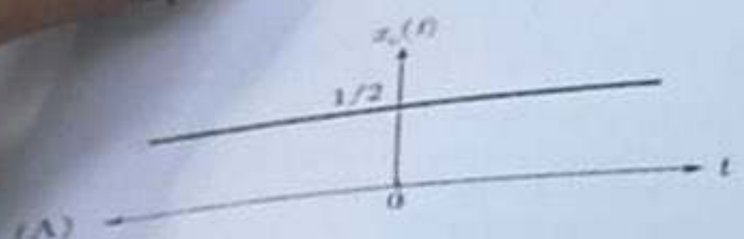
d. $y[n] =$ $x[n+1]$

..... system is a ..... system.

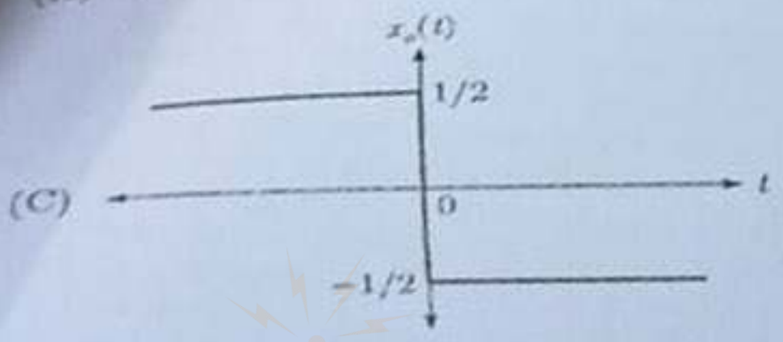
b. Time-invertible, causal. c. Stable, memoryless. d. Memoryless, causal e. Causal, stable

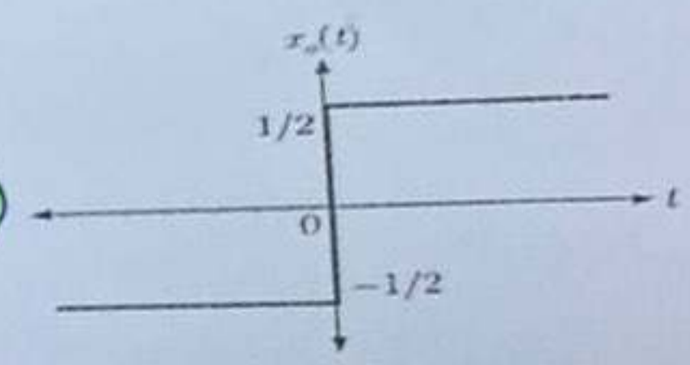
part of a unit step signal is.....

(A)

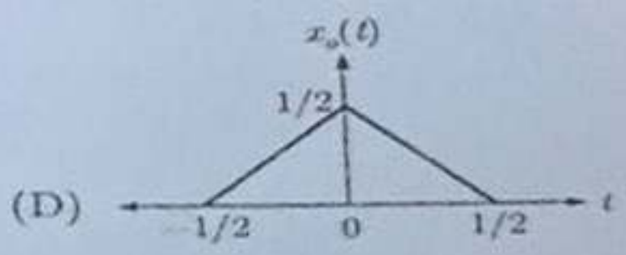


(C)



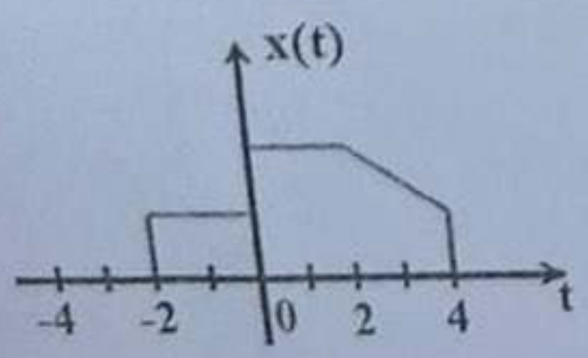


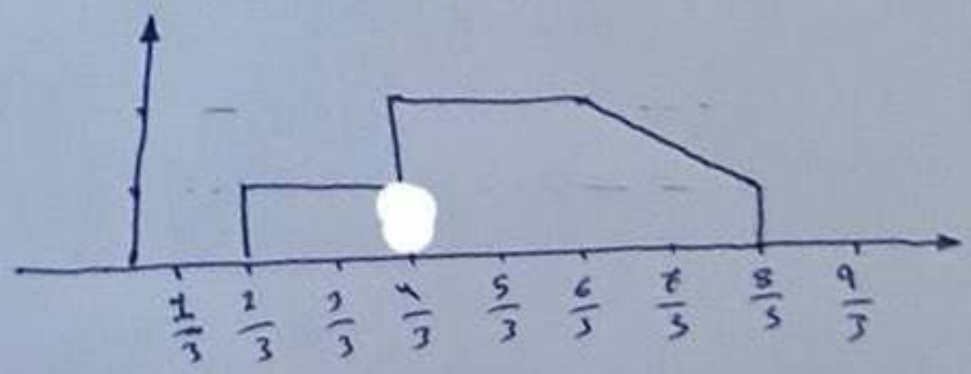
(D)



Q2. Answer the following:

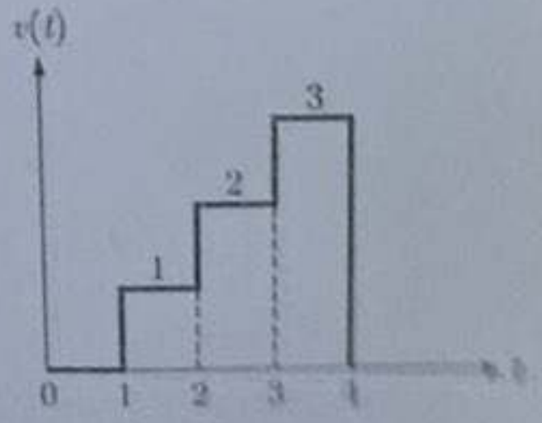
2.1. For the signal $x(t)$ shown below, sketch $x(3t-4)$.





2. Express the following figure in a unit step form as.....

$u(t-1) + u(t-2) + u(t-3) - 3u(t-4)$



Good Luck!